

DEY EDUCATION

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[M.Sc., M.Tech., Ph.D, OVDF]

Gravitation Problem Set

1. Write Newton's law of gravitation. What is the gravitational constant? Write its value in SI and CGS unit.
2. What is the gravitational constant ? Write its unit in SI and CGS. Write its expression.
3. What should be the change in g due to change in height and depth – find the expression.
4. What should be the change in g due to rotation of Earth? Here, the weight of an object is maximum on Earth's surface.
5. Write three laws of Kepler. Prove the second and third law.
6. What is escape velocity and determine its expression.
7. Why light gases are not available in the Earth's atmosphere?
8. Prove that, the Moon will not rotate the Earth if the velocity of the Moon increased by 42%.
9. Why a person will not feel any weight in an artificial satellite?

10. A person is able to jump 2m on Earth's surface. What should be the jumping height of the person on other planet (density of the planet is $\frac{1}{3}$ of Earth's and the radius is $\frac{1}{4}$ of Earth) ?
11. The value of gravitational acceleration is same at a height h and depth x . Find the relation between x and h .
12. Find the potential and kinetic energy of Moon. From this, find total energy of Moon. Write the significance of '-' sign.
13. Mass of the Moon is $\frac{1}{80}$ th of the mass of the Earth, and the diameter of the Moon is $\frac{1}{4}$ th of the diameter of the Earth. Find the ratio of gravitational acceleration of Moon and Earth surface.
14. Due to the Earth's diurnal motion, determine the difference between gravitational acceleration at the pole and at line of sight.
15. An object of mass 10 kg is thrown vertically upward with a velocity of 2 m/s from the top of a tower. The height of the tower is 10 m. Find the kinetic energy of the object at the moment it touches the ground.
16. A lift is moving upward with acceleration f . A person standing in the lift throws a ball with velocity v with respect to the lift in vertically upward direction. After time t , the ball returns to the person. Show that $v = (f + g) \frac{t}{2}$.

17. A balloon is at rest. A stone is left from the balloon at the last $\frac{1}{4}$ sec of motion. The stone travels 81 ft and then touches the ground. Determine the height of the balloon and velocity of the stone when it touches the ground.
18. A stone is left from the top of a tower of height 400 m. Another stone was thrown from the bottom of the tower at the same instant with velocity of 100 m/sec in vertically upward direction. When and where these two stones will meet each other. ($g = 9.8 \text{ m/s}^2$).
19. The length of a pendulum is increased by 21%. At what percentage the time period will increase ?
20. Show that if the velocity of the Moon is increased by 42%, then it will not continue to orbit around the Earth.
21. Two objects of mass m and M are at infinity from each other. Due to the gravitational force, these two objects are coming closer to each other. When the separation of the two objects is r , prove that, at that moment the relative velocity is $\sqrt{\frac{2G}{r}(M + m)}$.
22. A lift was moving upward with an acceleration of 2 m/s^2 . When the velocity of the lift is 8 m/sec, at that time a nail falls from the top of the lift. If the height of the lift is 3 m then, after what time the nail will touch the ground of the lift ?
23. The radius of the Earth is R . The gravitational acceleration at a height h and at a depth h are equal. Show that $h = \frac{1}{2}(\sqrt{5} - 1) R$.

24. Show that, the distance travelled by an object thrown vertically upward at the last second of the motion is equal to the distance travelled by a free-falling body at the first second of its motion.
25. Two pendulums have time periods of 1.8 sec and 2 sec, started to oscillate at the same time. After what time the faster pendulum will make one extra oscillation than the other one ?
26. If the length of a pendulum is increased by x , the time period will change from T_1 to T_2 . Prove that $g = \frac{4\pi^2 x}{T_2^2 - T_1^2}$
27. A seconds pendulum shows the correct time at one place on the Earth's surface, but it slows 20 sec (everyday) when it was taken to the peak of a mountain. Find the height of the mountain.
28. A satellite is at a height of 300 km from the Earth's surface, rotating around the Earth in a spherical orbit. Find the velocity of the satellite. (Radius of Earth = 6400 km and $g = 980 \text{ cm/sec}^2$).
29. An object is released from a distance r (measured from the centre of the Earth). If the radius of the Earth is R , then show that it will touch the ground with a velocity of $R \left[2g \left(\frac{1}{R} - \frac{1}{r} \right) \right]^{\frac{1}{2}}$
30. If the distance of the Earth from the Sun became half of its present, then how many days will be in a year?